Rule-Based AI Chatbot Project Report

Internship Project 1 – Artificial Intelligence

Student Name: Sandhya Raghav

Project Objective

The objective of this project is to design and implement a simple rule-based AI chatbot using Python. The chatbot interacts with users through predefined responses and demonstrates the fundamentals of Artificial Intelligence and decision-making logic.

Tools & Technologies Used

• Python Programming Language
• Visual Studio Code (VS Code)
• Command Prompt / Terminal

Project Introduction

A rule-based chatbot is one of the simplest forms of Artificial Intelligence. It follows predefined rules and responds according to conditions written by the programmer. This project helps beginners understand AI concepts, loops, conditional statements, and user interaction.

Key Features

• Handles greetings such as Hello, Hi, and Hey
• Answers common questions
• Responds to thank-you messages
• Supports exit commands (Bye, Exit, Quit)
• Runs continuously until the user exits
• Easy to understand and beginner-friendly

Working Procedure

1. Program starts and displays a welcome message.
2. User enters a message.
3. Input is converted to lowercase.
4. Conditions are checked using if-elif-else statements.
5. Appropriate response is displayed.
6. Program continues until the user types bye, exit, or quit.

Python Source Code

```
print("Welcome to AI Chatbot!") while True: user = input("You: ").lower() if user in
["hello", "hi", "hey"]: print("Bot: Hello! Nice to meet you.") elif user == "how are
you": print("Bot: I am doing great!") elif user == "what is your name": print("Bot:
My name is RuleBot.") elif user == "who made you": print("Bot: I was created by an
AI Intern.") elif user == "what can you do": print("Bot: I can answer simple
questions.") elif user == "thank you": print("Bot: You're welcome!") elif user in
["bye", "exit", "quit"]: print("Bot: Goodbye!") break else: print("Bot: Sorry, I
don't understand that.")
```

Sample Output

You: hello
Bot: Hello! Nice to meet you.

You: what is your name
Bot: My name is RuleBot.

You: bye
Bot: Goodbye!

Advantages

• Easy to develop and understand
• Fast response time
• Suitable for beginners learning AI
• Requires minimal resources

Future Improvements

The chatbot can be enhanced using Natural Language Processing (NLP), voice recognition, machine learning, and database integration to support more intelligent conversations.

Conclusion

This project successfully demonstrates the basic concepts of Artificial Intelligence through rule-based decision making. The chatbot uses Python control flow and conditional logic to simulate human interaction and provides a strong foundation for advanced AI projects.

Output